

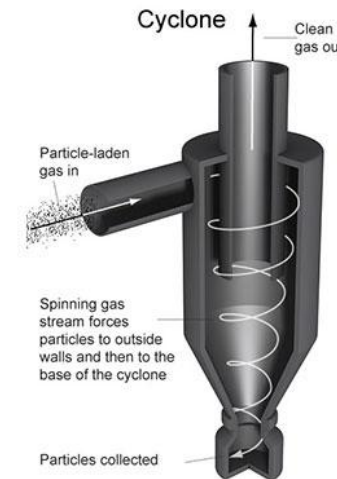
Sound = pressure wave

Filters

Coffee filter

Particle filter (to measure pollutions)

Filters in Signal Processing (remove or attenuate some frequencies while keeping the others without any change)

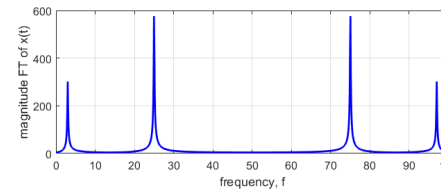
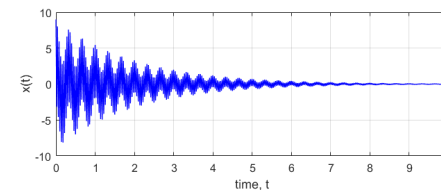
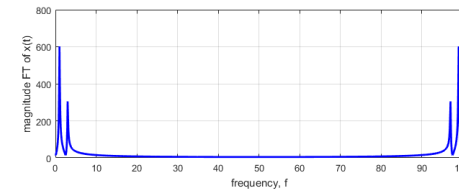
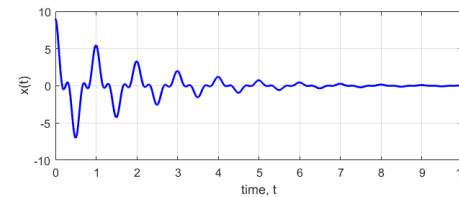


Recall (Lecture 3)

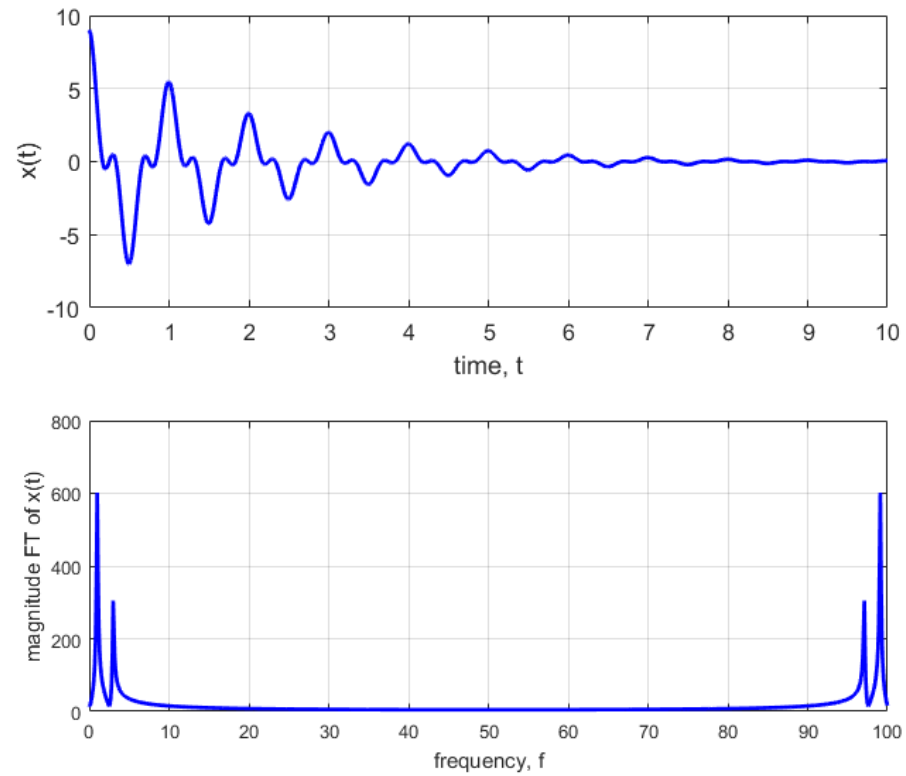
A signal can be presented as a mixture of sines and cosines (complex exponentials)

$$\int_{-\infty}^{+\infty} X(f) e^{-j2\pi f t} df = x(t)$$

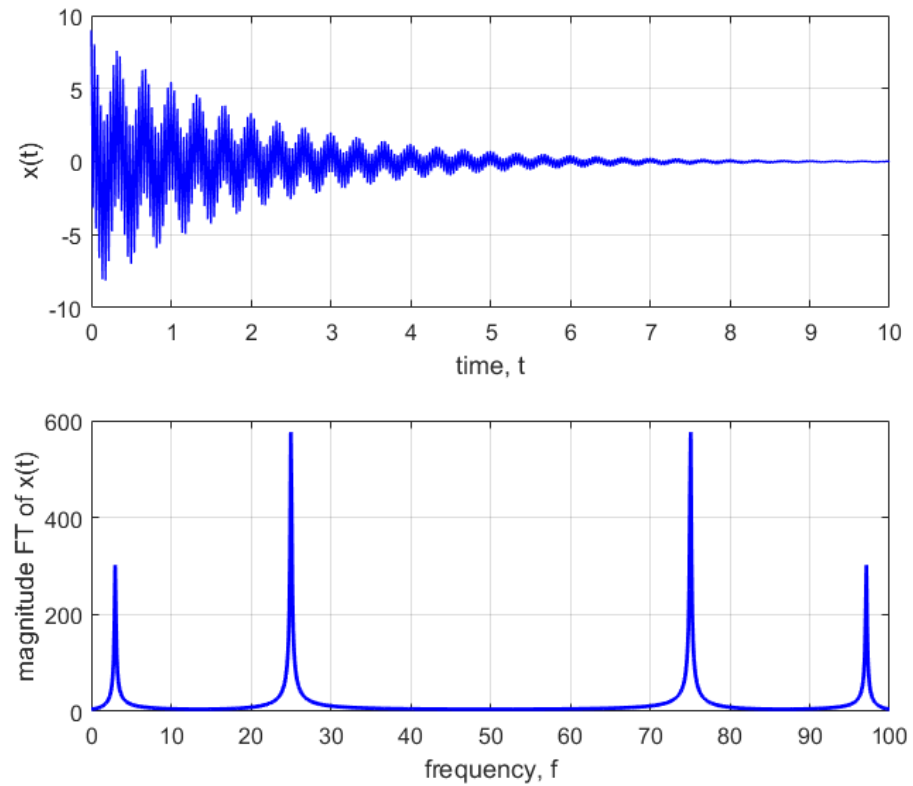
$X(f)$ tell us how much each frequency contributes



Illustration



Illustration



How are filters applied?

$$x(t) \xleftrightarrow{FT} X(f)$$

Filter response $h(t)$ to a narrow pulse on the input (think of it as a delta-function)

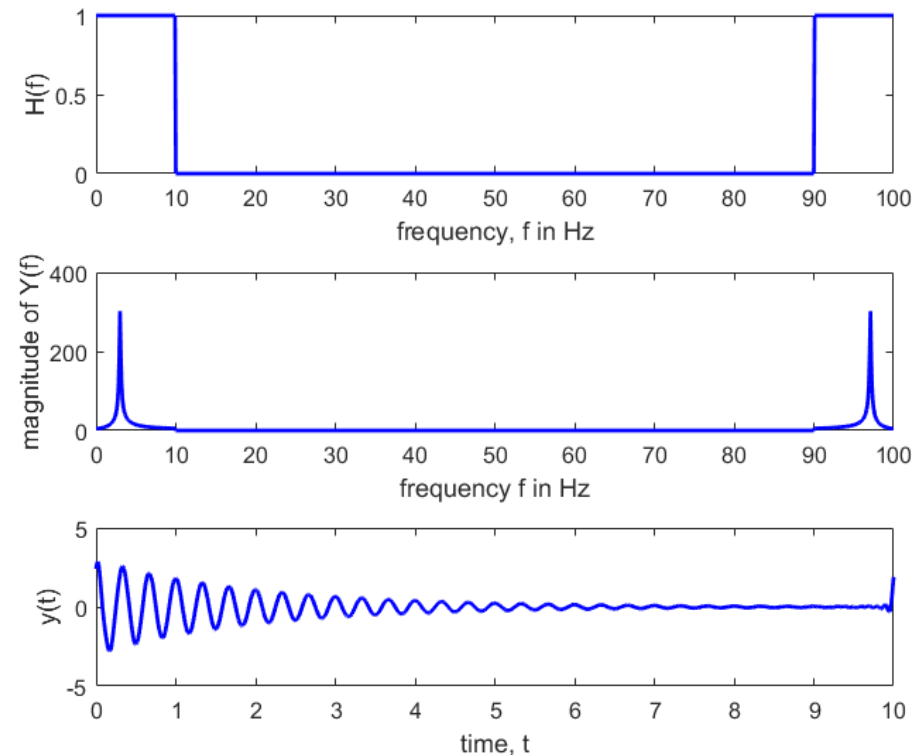
$$h(t) \xleftrightarrow{FT} H(f)$$

The response of filter on a composed input $x(t)$ is a correlation of $x(t)$ and $h(-t)$

By Correlation Theorem:

$$Y(f) = X(f)H(f)$$

$y(t)$ is the output of the filter.



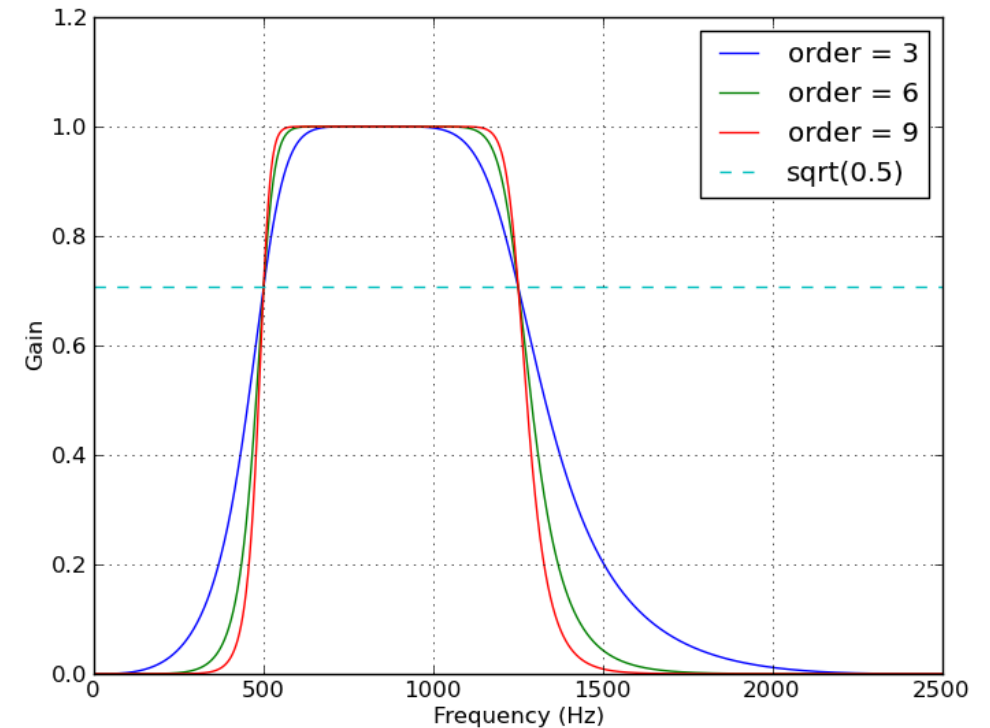
Four types of filters

Low pass = suppresses high frequencies,
passes low frequencies

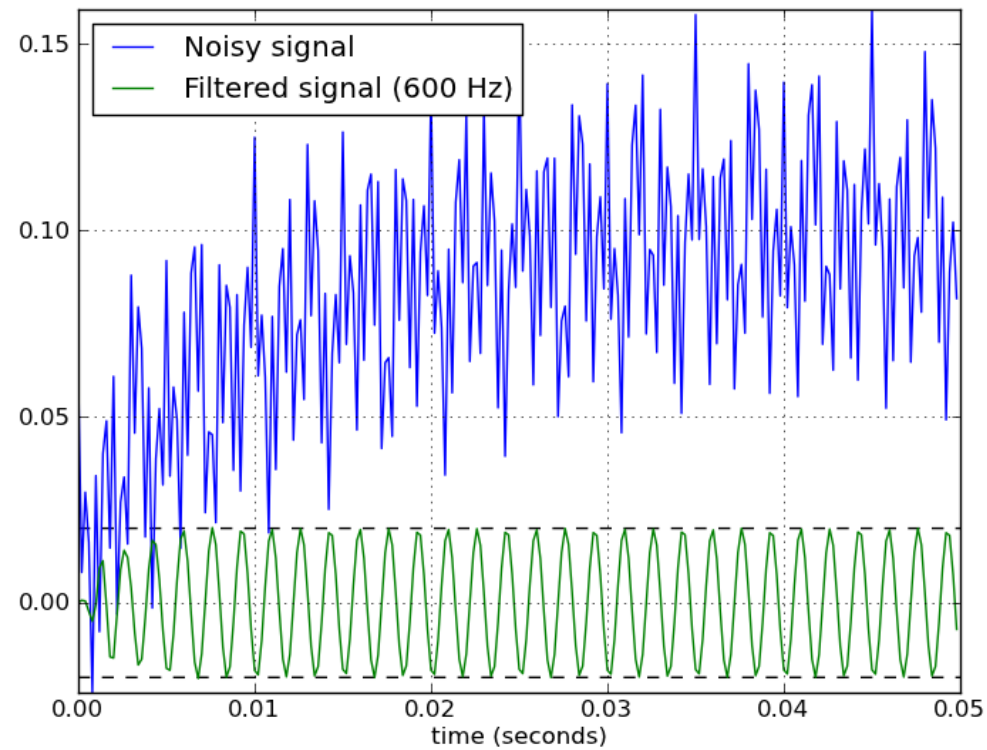
High pass = passes high frequencies and
suppresses low

Band pass = passes medium frequencies,
suppresses low and high

Band stop



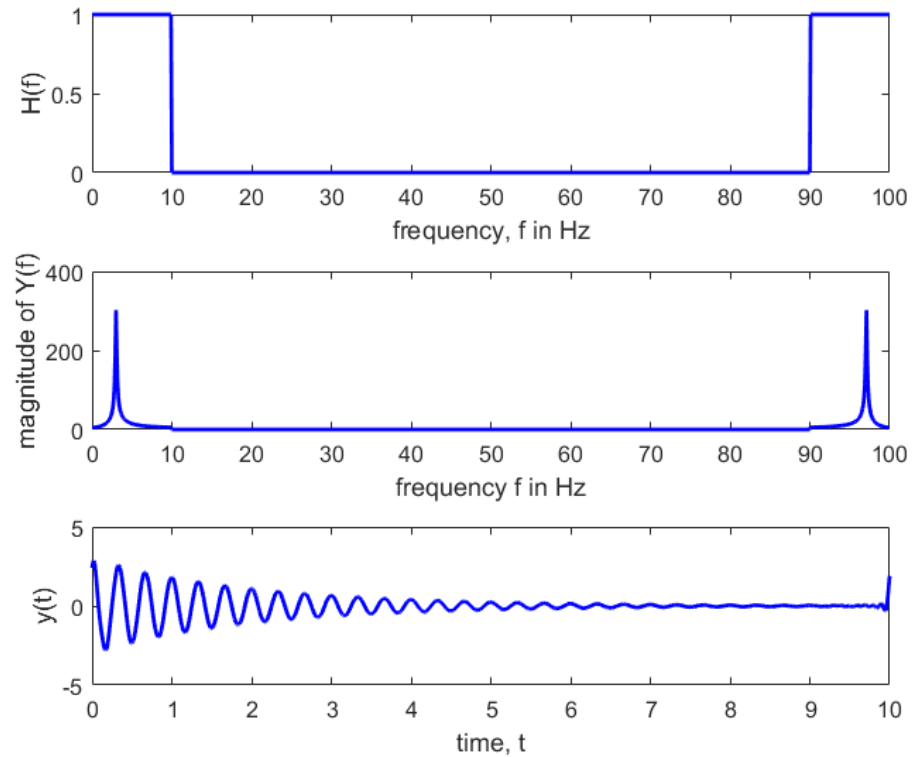
Illustration



Filters

Low pass
filtering: the
original signal
and its FT

Signal and its FT after low pass filtering



Filtering in 2D

Simulated pulse in filter-bank format

(image)

DM = 375

S/N = 25

Pulse width $T = 50$ ms

Apply low-pass filter in 2d FT space

